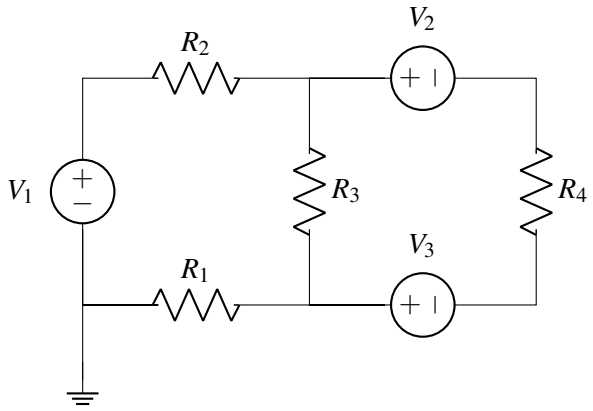
CSM
Fall 2026

EE 64
Worksheet 3

1. Supernode

In this question, we will explore how to deal with multiple voltage sources when doing nodal analysis. Throughout the problem, assume that $R_1 = 1\Omega, R_2 = 2\Omega, R_3 = 4\Omega, R_4 = 4\Omega, V_1 = 1\text{V}, V_2 = 2\text{V}, V_3 = 4\text{V}$. While it is good practice to symbolically solve problems before substituting specific numbers, for superposition it is often easiest to substitute before superposing the solwers.



(a) Mark all the nodes. If you know the potential at the node, write down the value next to the node. If you don't know the value, then assign a variable for the potential.

(b) Mark current directions arbitrarily and corresponding polarities on each resistor. Note that if current goes from left to right, then the left side of the resistor is to be marked + and the right side must be marked -. This is the passive sign convention.

(c) Note that we define 4 nodes with unknown potentials. So we need 4 equations. Each of these nodes with unknown potential should give us one equation.
Write the equation for the first node with unknown potential.

(d) Write equations for all remaining unknown nodes. Solve for all the unknown potentials.

(e) Now we will explore solving the circuit using superposition. To recall the process, which 2 sources will be suppressed when looking at each of the 3 sources. Will the other 2 sources be treated as open circuits or short circuits?

(f) Draw the superposition circuit for V_1 and solve this circuit.

(g) Draw the superposition circuit for V_2 and write the equation that characterizes this circuit.

(h) Draw the superposition circuit for V_3 and write the equation that characterizes this circuit.

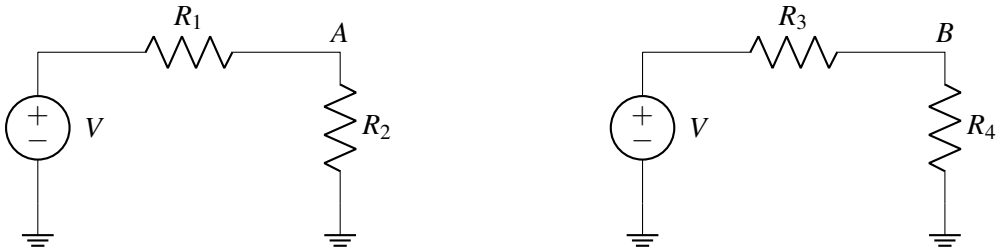
(i) Find the unknown potentials as you did in in part (d), but now do so using the superposition theorem.

CSM, Fall 2026. All Rights Reserved. This may not be publicly shared without explicit permission.

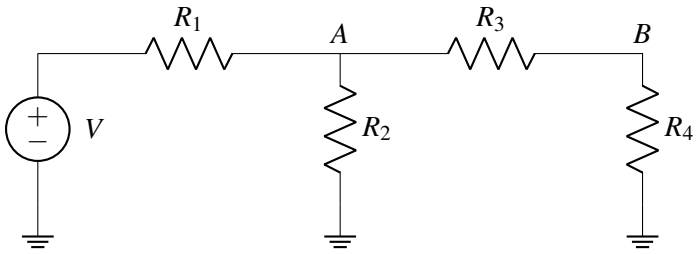
2. Modular Circuits

In this question we will explore how to look at circuits in a 'modular' way. That is, we will start to encapsulate circuit components (just like one might encapsulate code into functions) and then see how those components fit together.

(a) Here, you can see two standard voltage dividers. Find the voltage at the points A and B using standard nodal analysis techniques.



(b) Now, let us modify this circuit to add a second voltage divider stage starting at A. You can think of this as cascading 2 voltage dividers: one which which has an input of V and an output of V_A, and a second one that has an input of V_A and an output of V_B. Find the voltage at B using nodal analysis.



(c) Compare your solwer with what you get when you multiplying the amplification factors of each individual voltage divider. Are they the same or different? Does this surprise you? Why or why not?

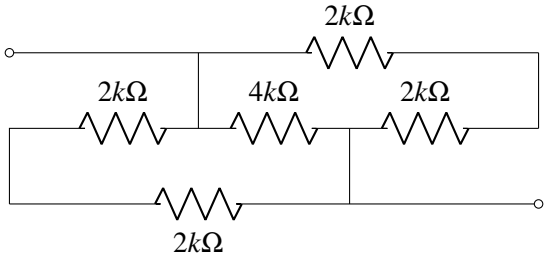
CSM, Fall 2026. All Rights Reserved. This may not be publicly shared without explicit permission.

3. Never fail at resistor equivalence

(a) Using what you learned about resistance ($R = \frac{\rho \cdot l}{A}$), explain what would happen if you connect two resistors in series.

(b) Using the same resistance equation from part (a), explain what would happen if you connect two resistors in parallel.

(c) Now let's apply some of these ideas to a circuit. For the circuit below, mark all the nodes.



(d) Mark which nodes are 2-nodes and multi-nodes. 2 nodes are connected to only 2 components, and multi-nodes are connected to 3 or more components.
Note: *Whenever nodes are marked across which equivalent resistance must be found, those are considered 'components' because something could be connected there.*

(e) Resistors that are connected to 2-nodes are considered to be in series. Redraw the circuit, and find the 2 and multi-nodes again after combining the resistors connected to 2 nodes.

(f) Now we should be left with only multi-nodes. So far, we have seen that any resistors connected to 2-nodes are in series. We will see what happens to resistors connected to multi-nodes. Begin by writing out the 2 nodes that each resistor is connected to.

(g) If you have 2 or more resistors that are connected to the same 2 nodes, then they are in parallel. What does this mean for the 3 remaining resistors?

CSM, Fall 2026. All Rights Reserved. This may not be publicly shared without explicit permission.